Research Highlights

Manuscript: *A Layered, Axiom-Isolated Lean 4 Formalization of the Congruent Number Problem up to Tunnell's Criterion*

- 954-line Lean 4 formalization with zero sorry placeholders
- Axiom isolation pattern separating BSD from proved content
- Formal non-vacuity proof for the axiom boundary
- Fixed-point-free Finset involution proved by induction
- Python pipeline validates criterion for squarefree $n \leq 500$